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Algorithmic Bias in Education: Project Proposal

Introduction and Motivations

Technological and computational algorithms, in many of the contexts in which they are becoming increasingly prominent, are steeped with bias, and are influencing educational systems where they are utilized. Prediction models for student success in and beyond primary and secondary education have the potential to show unfair bias towards students from different backgrounds. Opportunity for students should not be hindered based on attributes, including gender or race, classified by an algorithm. Data is collected for these models in ways that are not conducive to equitable results; “if we do not collect data from the right sample of learners, we cannot expect our models to work on all learners” (Baker and Hawn 1064). This can create algorithms that reinforce preconceived notions about students, and can limit their education opportunities and expectations, but to what extent? We will examine three aspects of data originating from algorithms in the educational sphere:

1. How are exam score predictions impacted by factors like a student's socioeconomic status, race, gender, or disability status for example?
2. Is placement in accelerated courses, such as AP/IB for high school students, influenced by collected information regarding the background of a student?
3. How are future “success” trajectories predicted for elementary and secondary students, specifically in terms of higher education and work force predictors?

Methods

For preliminary processing, variables with more than 75% missing values will be removed. Additionally, we will aggregate and filter data using Python and SQL as necessary for computational clarity.

Exploratory data analysis will be conducted to understand the structure of the data. Data will be disaggregated during this process so that characteristics of both individual schools and

larger geographic areas can be taken into account when considering statistics regarding exam scores, course placement, and outcome after graduation.

Various regression methods will be used to analyze our research questions, including but not limited to multiple linear regression and logistic regression to examine relationships in data, and hierarchical regression models to determine relative feature importance.

Data

Two data sources will be primarily used for this project: the US Department of Education's (ED) Civil Rights Data Collection (CRDC), and the High School Longitudinal Study (HSLS), conducted by the ED, Institute of Education Sciences, and National Center for Education Statistics.

The CRDC is a large collection of data collected from schools and school districts across the United States. The main purpose of this data is to help inform policy to combat discrimination on the basis of race, nationality, sex, and disability in educational institutions which receive Federal financial assistance from the ED. This dataset contains information about students' and schools' demographic characteristics, as well as their academic lives, ranging from course placement to social climate.

The HSLS similarly contains data on students and their educations from across the United States, but is aimed towards understanding students' trajectories through secondary and postsecondary education. This dataset also contains information about possible influences for educational choices, including motivations to study certain subjects and influence from families.

We will be using variables from both datasets containing information about individual student demographics, such as race, gender, socioeconomic status, disability, etc., as well as aggregated school demographics, school characteristics, and information pertaining to educational outcomes such as course placement and exam scores on both student and institution levels.

Research Expectations

Overall, we expect that factors like socioeconomic status, race, gender, disability status, etc., will noticeably impact educational outcomes, with affluent, white, able-bodied, male students having the highest projected outcomes across all research questions.

Hypothesis 1: We hypothesize that demographic features of students, as well as their disability status, would have an impact on the predictions of test scores such as SAT and AP tests. Predictors will likely assign higher test scores to students without disability status and higher family income for example.

Hypothesis 2: In addition to the demographic factors above, we predict that geographic location of school, type of school, and general demographics of school and school area would be meaningful predictors of a student's placement in advanced coursework, as it is highly dependent on the school's offerings. Specifically, we hypothesize that schools with higher funding in urban and suburban areas would more likely have higher proportions of students placed in advanced coursework.

Hypothesis 3: We hypothesize that demographic factors, socioeconomic factors, and school characteristics will influence predictions of students' future educational and workplace success. We predict that students from higher income families and schools with greater resources will be more likely to be predicted to pursue higher education and have more successful outcomes in the workforce.

Group Contract and Work Agreement

Our group will communicate openly about our individual responsibilities, availability, and any conflicts that may affect our work. We may not all be able to do the same amount of work in a given time period, but we will communicate with each other early on if so. We can acknowledge our different academic backgrounds by dividing tasks based on each other's previous coursework, skill sets, and interests while still making sure we all are contributing meaningfully. At a minimum, we have agreed to meet weekly on Tuesdays at 4:30 p.m. and schedule additional meetings as needed based on workload and deadlines. We will keep track of our individual contributions through Git and clearly annotate commits to show what was changed or contributed. While the amount of work between us may vary at times, we will share regularly to make sure responsibilities are distributed fairly. We have talked about specific days and times of the week when we won't be available based on other commitments (Grace writes

and edits for the Misc, Maia performs in the orchestra, and Amanda is on the lacrosse team). Overall, we agreed that open communication is the most effective way to ensure we finish all of our work on time and at a high quality.

References

Baker, R.S., Hawn, A. “Algorithmic Bias in Education”. *International Journal of Artificial Intelligence in Education*, 32, 2021, 1052–1092.

Bird, K. A., Castleman, B. L., & Song, Y. “Are algorithms biased in education? Exploring racial bias in predicting community college student success”. *Journal of Policy Analysis and Management*, 44, 2024, 379–402.